Federico Chinello

Phone: +39 339 1227382 | Email: federico.chinello@studbocconi.it | Personal Website | Google Scholar | GitHub
Birth date: 31/10/1998 | Nationality: Italian

Education

9/2023–4/2026 **M.Sc. in Artificial Intelligence, Bocconi University | Final grade: 110/110 cum laude**
Thesis: Convolutional Set Transformer
Cutting-edge Computer Science degree. GPA: 30.02/30.

9/2020–3/2023 **M.Sc. in Finance, Bocconi University | Final grade: 110/110 cum laude**
Thesis: A look inside the "black box": investors' behaviour and price formation in the closing auction
Focus on Computational Finance and analysis of financial Big Data, deepened through internships.

9/2017–10/2020 **B.Sc. in Economics and Finance, Bocconi University | Final grade: 110/110 cum laude**
Thesis: Equilibrium price and volume discovery in a single price batch auction: an empirical analysis
Focus on Computational Finance and analysis of financial Big Data, deepened through internships.

9/2012–7/2017 **Maturità classica (classical studies)**
Liceo Galileo Galilei, Legnano, Italy

Professional Experience

10/2025-present **Research Fellow, AIRC Institute of Molecular Oncology (IFOM), Milan**
Developing computer vision and AI solutions to support biomedical research.

6/2025-9/2025 **Research Assistant, Department of Computing Sciences, Bocconi University, Milan**
Supervisors: Prof. Francesca Buffa and Dr. Ylli Doksani (IFOM PI)
Developed software to automate the analysis of electron microscopy images of DNA molecules.

9/2024-11/2024 **Research Assistant, Department of Computing Sciences, Bocconi University, Milan**
Supervisor: Prof. Francesca Buffa
Contributed to a project focused on enabling robust and reproducible evaluation of Machine Learning models in biomedical sciences.

7/2022–6/2023 **Pre-Doctoral Fellow, Algorand Fintech Lab, Bocconi University, Milan**
Supervisor: Prof. Barbara Rindi
Developed robust Python pipelines for large-scale processing and analysis of blockchain data. Reviewed smart contract code and contributed to theoretical research and analysis.

2/2020–9/2021 **Research Assistant, IGIER, Bocconi University, Milan**
Supervisor: Prof. Barbara Rindi
During three internships, developed highly efficient Python and MATLAB pipelines to process tens of terabytes of high-frequency stock trading data (nanosecond resolution).

Honors, Awards & Fellowships

2025 **Research fellowship** AIRC Institute of Molecular Oncology

2022 **Pre-Doctoral fellowship** Algorand Fintech Lab, Bocconi University

2017 **Merit-based scholarship** Fondazione Famiglia Legnanese

Publications

Convolutional Set Transformer pre-print | code
Chinello, F., & Boracchi, G., arXiv preprint, 2025.
We introduce a novel neural architecture designed to process image sets of arbitrary cardinality that are visually heterogeneous yet share high-level semantics (such as a common category or concept).

Software

DNA2Graph code | website | PyPI
Imaging software that automates the segmentation and analysis of DNA molecules in electron microscopy images, streamlining workflows that were previously done manually by researchers.

cstmodels code | website | PyPI
This package provides the reference implementation of the Convolutional Set Transformer (Chinello and Boracchi, 2025). It includes reusable Keras 3 layers for building CST architectures, and provides an easy interface to load and use the CST-15 model pre-trained on ImageNet.

Computer skills

C/C++ proficient | **Python** proficient | **SQL** proficient | **SLURM** proficient | **Git** proficient
MATLAB intermediate | **Solidity** intermediate
AFL (fuzzer) basic | **GNU MathProg** basic | **GLPK** basic

Languages

Italian native | **English** proficient | **Spanish** intermediate | **Latin** | **Ancient Greek**

Volunteering

5/2022–current **Politics Hub APS, Legnano (Milan)**
We organize talks with leaders in politics, business, and academic research. Board member in 2023.

[*I hereby authorize the use of my personal data in accordance to the local low*]

Milan - Italy, May 27, 2026